

Exp No:20 Experiment to understand K-means clustering algorithm for a given data set.

Description: Understand the K-means clustering algorithm for the dataset given.

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
```

```
In [2]: df=pd.read_csv('Mall_Customers.csv')
```

```
In [3]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   CustomerID                           200 non-null   int64
1   Gender                               200 non-null   object
2   Age                                   200 non-null   int64
3   Annual Income (k$)                   200 non-null   int64
4   Spending Score (1-100)                200 non-null   int64
dtypes: int64(4), object(1)
memory usage: 7.9+ KB
```

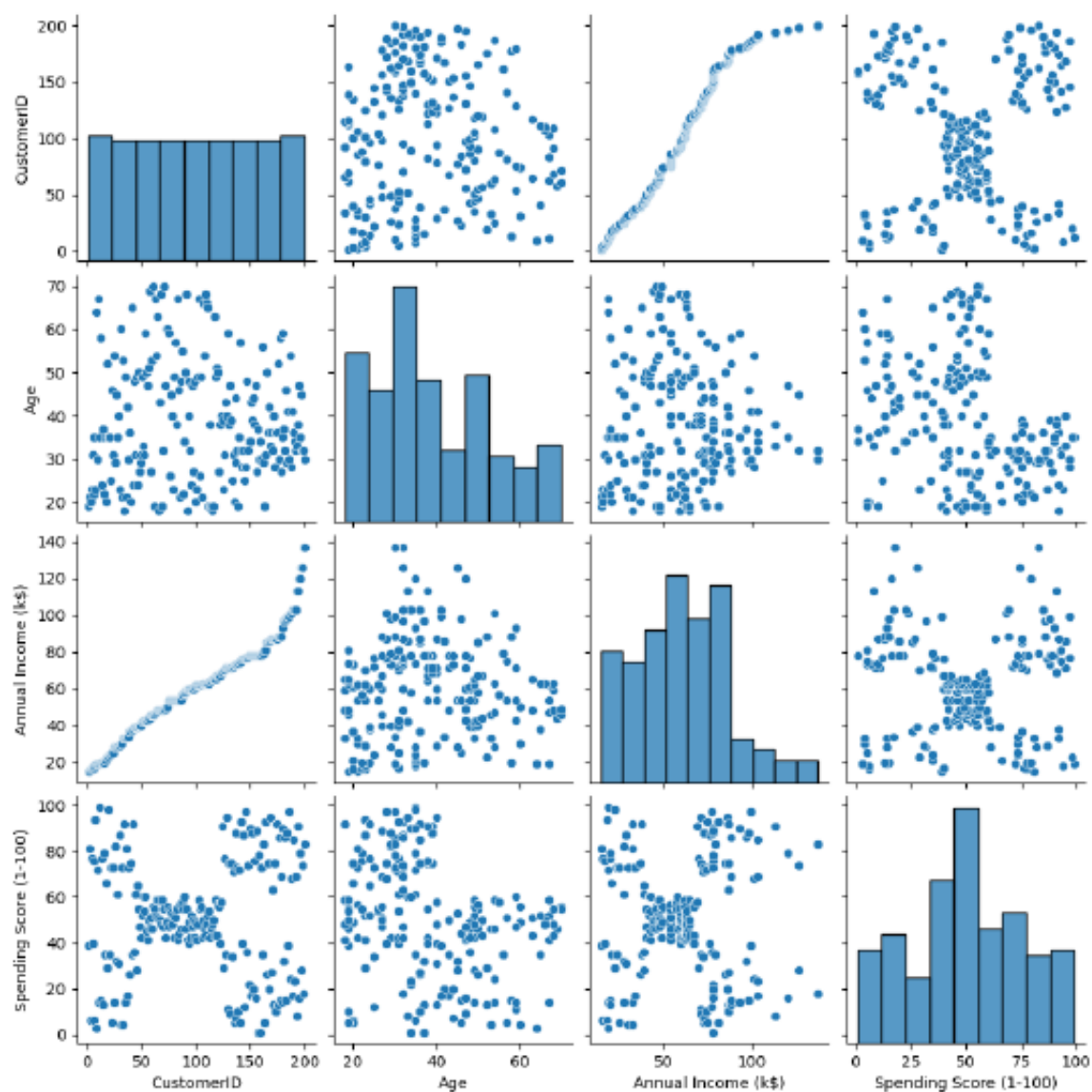
```
In [4]: df.head()
```

```
Out[4]:
```

	CustomerID	Gender	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

```
In [5]: sns.pairplot(df)
```

```
Out[5]: <seaborn.axisgrid.PairGrid at 0x170e8e47850>
```



```
In [6]: features=df.iloc[:,[3,4]].values
```

```
In [7]: from sklearn.cluster import KMeans
model=KMeans(n_clusters=5)
model.fit(features)
KMeans(n_clusters=5)
```

C:\Users\Ayyadurai\AppData\Local\anaconda3\Lib\site-packages\sklearn\cluster_kmeans.py:870: FutureWarning: The default value of `n\_init` will change from 10 to 'auto' in 1.4. Set the value of `n\_init` explicitly to suppress the warning

warnings.warn(
C:\Users\Ayyadurai\AppData\Local\anaconda3\Lib\site-packages\sklearn\cluster_kmeans.py:1382: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=1.
warnings.warn(
)

Out[7]: KMeans(n_clusters=5)

In a Jupyter environment, please rerun this cell to show the HTML representation or trust the notebook.

On GitHub, the HTML representation is unable to render, please try loading this page with nbviewer.org.

```
In [8]: Final=df.iloc[:,[3,4]]
Final['label']=model.predict(features)
Final.head()
```

C:\Users\Ayyadurai\AppData\Local\Temp\ipykernel_8116\470183701.py:2: SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

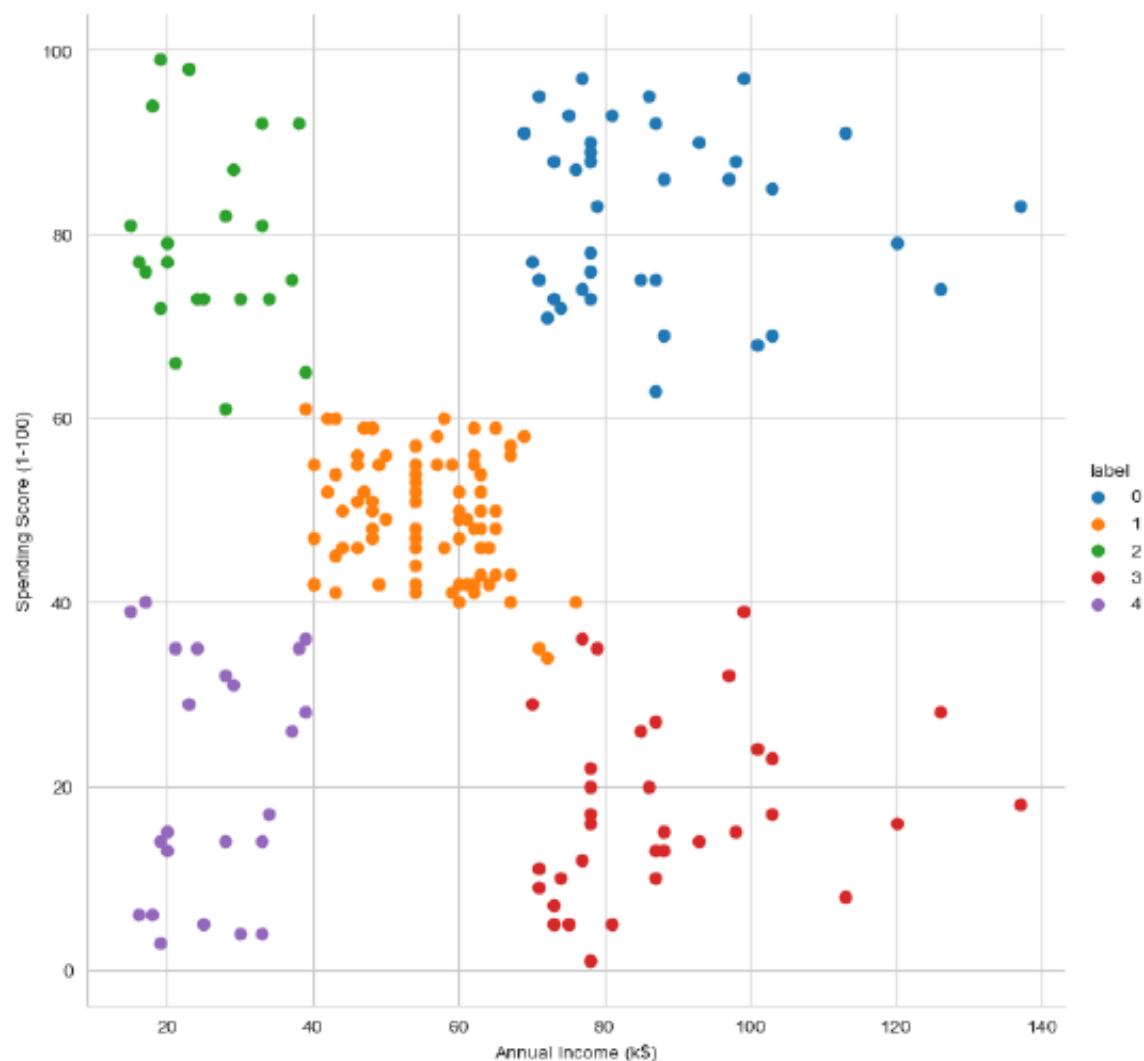
See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy (https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy)

```
Final['label']=model.predict(features)
```

Out[8]:

	Annual Income (k\$)	Spending Score (1-100)	label
0	15	39	4
1	15	81	2
2	16	6	4
3	16	77	2
4	17	40	4

```
In [9]: sns.set_style("whitegrid")
sns.FacetGrid(Final, hue="label", height=8) \
.map(plt.scatter, "Annual Income (k$)", "Spending Score (1-100)") \
.add_legend();
plt.show()
```

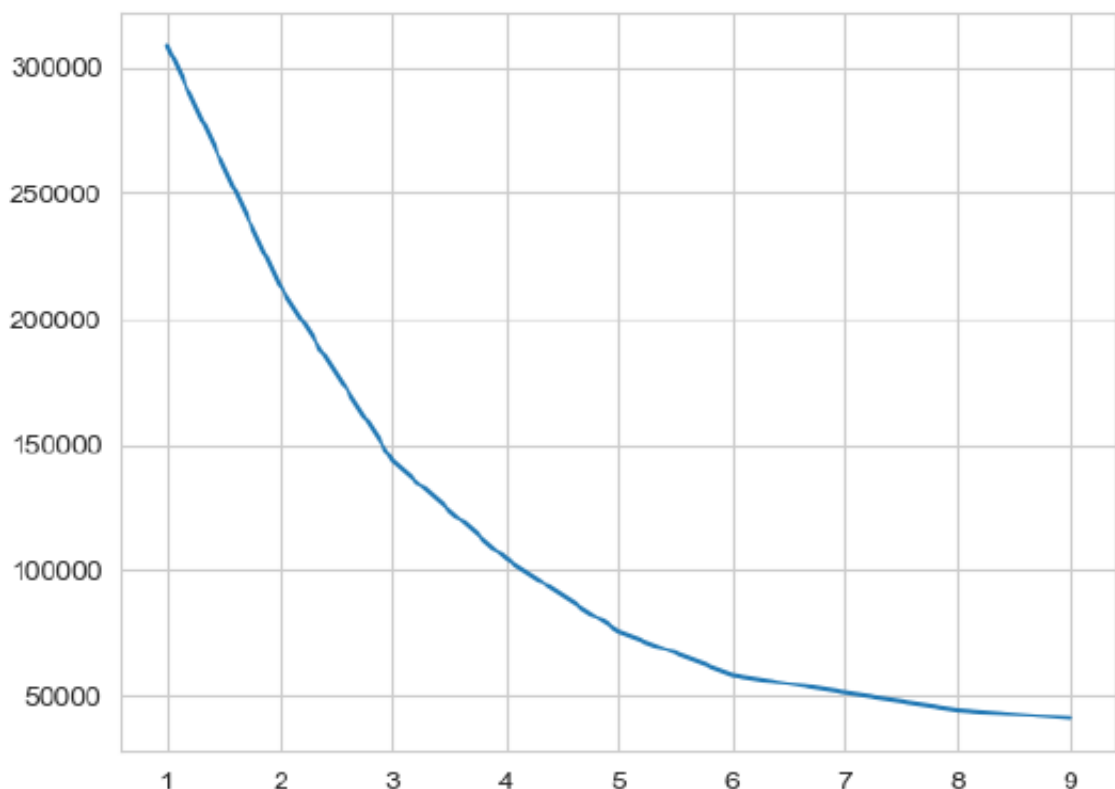


```
In [10]: features_el=df.iloc[:,[2,3,4]].values
from sklearn.cluster import KMeans
wcss=[]
for i in range(1,10):
    model=KMeans(n_clusters=i)
    model.fit(features_el)
    wcss.append(model.inertia_)
plt.plot(range(1,10),wcss)
```

s/K-Means Clustering.ipynb

```
er\_kmeans.py:870: FutureWarning: The default value of `n_init` will change from 10 to 'auto' in 1.4. Set the value of `n_init` explicitly to suppress the warning
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C:\Users\Ayyadurai\AppData\Local\anaconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1382: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=1.
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  warnings.warn(
```

Out[10]: [<matplotlib.lines.Line2D at 0x170e99f3550>]



In []:

